

Adarsh Bhandary Panambur

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Professional Experience

Research Scientist, National High Performance Computing Center (NHR@FAU), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) 06/2026 – Present

- Conduct research in Artificial Intelligence (AI) and High-Performance Computing (HPC).
- Develop, deploy, and optimize **Large Language Models (LLMs)**, **Vision-Language Models (VLMs)**, and multimodal AI systems.
- Design scalable training and inference pipelines on **GPU clusters** and **supercomputing infrastructure**.
- Build **AI-powered applications**, agentic workflows, and research software using foundation model APIs and **open-source AI** frameworks.
- Collaborate with **academic and industrial partners** to translate AI research into practical solutions.

PhD Researcher, Friedrich-Alexander-University Erlangen-Nuremberg 05/2020 – Present | Erlangen, Germany (Pattern Recognition Lab)

Thesis: Deep learning algorithms for advanced mammography and breast cancer analysis

- Improved downstream breast cancer classification and detection using **SSL methods in mammography and contrast-enhanced mammography**.
- Sourced and curated multiple public datasets from multiple countries to develop a **comprehensive dataset repository** for mammography containing roughly **100,000 images**.
- Developed **multimodal image-text foundation models** using large vision-language models for enhanced **breast cancer** analysis and **assistive navigation** for the visually impaired.
- Ranked 4th in the **MICCAI ODELIA Challenge 2025** for **breast MRI malignancy** classification.
- **Supervised** multiple master's theses and research projects in collaboration with Siemens Healthineers, Siemens AG, and University Hospital Erlangen.
- Contributed to the co-writing of a multi-institution **research grant** focused on federated learning and foundation models.
- Developed **MLOps pipelines** with **Docker** and **SLURM** for automated training, experiment tracking, and reproducible deep learning models.

Guest Researcher, West China Hospital of Sichuan University 07/2024 – 09/2024 | Chengdu, China

- Performed **clinical validation** of Siemens Healthineers' **AI prototype for Breast MRI**, ensuring clinical applicability in cancer **screening and diagnosis**.
- Initiated **projects** to enhance breast cancer screening workflows.
- Streamlined **high quality data annotation** in collaboration with radiologists.

Doctoral Researcher, Siemens Healthineers 05/2020 – 02/2024 | Erlangen, Germany

- Performed **clinical validation** of **AI models for mammography** in an international, cross-department collaboration to enhance breast cancer screening in **Vietnam**.
- Led a project to **optimize hospital workflows** while ensuring regulatory compliance.
- Enhanced **mammogram data annotation quality** through collaboration with radiologists.
- Developed **DL methods** to categorize **positioning artifacts in mammography**.
- **Supervised research projects** through interdisciplinary collaborations.

Working Student & Master Thesis, Siemens Healthineers 10/2017 – 03/2020 | Erlangen, Germany

- Developed a **MATLAB-based software prototype** for the **prioritization** of tasks based on **compliance** requirements.
- Conducted detailed **prior art searches** and prepared multiple **successful claim charts**, directly supporting the legal IP team in assessing potential **patent infringements**.
- Evaluated **IoT technologies**, proposing new directions to optimize **clinical workflows**.
- Defined **software specifications** and documented **use cases for blockchain in healthcare**, co-inventing a method to unlock medical devices securely.
- Master's thesis on **deep learning**-based **lung nodule type** classification in **low-dose CT** using CNNs.

Senior Engineer (Medical Devices) 03/2015 – 02/2016 | Bangalore, India
Transcaal Engineers India Pvt. Limited

- Generated **25% of the company's annual revenue** through **sales of quality assurance** services while building and maintaining a hospital customer base across five Indian states.
- Led a **team of biomedical engineers** in on-site quality assurance projects while **hiring and mentoring new team members**.

Calibration Engineer (Medical Devices)

Transcaal Engineers India Pvt. Limited

• Conducted **testing and preventive maintenance of hospital equipment** using **Fluke biomedical test** devices, ensuring compliance with quality standards.

09/2014 – 02/2015 | Bangalore, India

Education

PhD Research Student

Friedrich-Alexander-University Erlangen-Nuremberg

Master of Science - Medical Data Engineering (Data and Image Processing)

Friedrich-Alexander-University Erlangen-Nuremberg

Bachelor of Engineering - Medical Electronics

Dr. Ambedkar Institute of Technology

05/2020 – Present | Erlangen, Germany
04/2016 – 03/2020 | Erlangen, Germany
08/2010 – 06/2014 | Bangalore, India

Skills

Programming Languages

Python, MATLAB, C++

Libraries and Frameworks

PyTorch, TensorFlow, Keras, Hugging Face, nnU-Net, OpenCV, MMDetection, MONAI, Albumentations, NumPy, pandas, Scikit-Learn, SciPy, XGBoost, SimpleITK, larvis, SLURM, Axolotl, DeepSpeed, SGLang

Data Modalities

Mammography, Chest X-ray, MRI, Low-dose CT, Natural images, Hematology, Industrial anomaly images, Sales data, Electrocardiogram

AI & ML

Self-supervised learning (SSL), Large vision language models (LVLMs), Classification, Segmentation, Detection

Version Control

Git, Docker

Languages, Hobbies and Interests

English

C2

Hindi

Native

Kannada

Native

German

B1

Konkani

Native

Hobbies and Interests

Cooking, Padel Tennis, Hiking, Table Tennis

Declaration

Adarsh Bhandary Panambur

Erlangen, 23.03.2026

Publications

First Author Publications

Conferences & Journals

- Adarsh Bhandary Panambur, Tri-Thien Nguyen, Saahil Islam et al. **BE-WISER: Class-Aware Weak Slice-Level Supervision for Breast MRI**, 02 June 2026, PREPRINT (Version 1) available at Research Square [<https://doi.org/10.21203/rs.3.rs-9855612/v1>]
- Panambur AB, Bhat S, Yu H, Madhu P, Bayer S, Maier A. **Attention-guided erasing for enhanced transfer learning in breast abnormality classification**. International Journal of Computer Assisted Rad. and Surgery. 2025 Jan 15:1-8. [↗](#)
- Bhandary Panambur, A., Bayer, S., & Maier, A. (2025). **Re-thinking Mammography Transfer Learning: The Dataset-Informed Transfer Learning (DITL) Framework for Breast Cancer Screening and Lesion Diagnosis**. In print at the International Conference on Computer Vision and Image Processing (2025).
- Bhandary Panambur, A., Wind, S., Bayer, S. & Maier, A., 2025. **MammoBLIP: End-to-End Mammography Report Generation with Vision-Language Models and Public Multi-Institutional Datasets**. In print at: IEEE Medical Imaging Conference (MIC) 2025, Yokohama, Japan. IEEE [↗](#)
- Panambur AB, Bhat S, Nguyen T, Bayer S, Maier A (2025). **Towards Foundational Models in Mammography: Leveraging Vision-Language Models for Radiological Reasoning**. Under internal review.
- Bhandary Panambur, A., Nguyen, T.-T., Bayer, S., & Maier, A. (2026). **BE-WISE: Breast MRI Evaluation with Weakly-Informed Slice-level Explanation**. Accepted at the German Conference on Medical Image Computing (BVM 2026).
- Bhandary Panambur, A., Yu, H., Bhat, S., Madhu, P., Bayer, S., & Maier, A. (2024). **Attention-Guided Erasing: A Novel Augmentation Method for Enhancing Downstream Breast Density Classification**. BVM 2024: Proceedings of the German Workshop on Medical Image Computing, Erlangen, Germany. [↗](#)
- Bhandary Panambur, A., Madhu, P., Bayer, S., & Maier, A. (2024). **Enhancing Downstream Classification of Breast Abnormalities in Contrast Enhanced Spectral Mammography using a Neighborhood Representation Loss**. Accepted at Medical Imaging 2024: Computer-Aided Diagnosis; SPIE. [↗](#)
- Panambur, Adarsh Bhandary, Prathmesh Madhu, and Andreas Maier. **Effect of Random Histogram Equalization on Breast Calcification Analysis Using DL**. BVM 2022: Proceedings, German Workshop on Medical Image Computing, Heidelberg. Wiesbaden: Springer Fachmedien Wiesbaden, 2022. [↗](#)
- Bhandary Panambur, A., Fieselmann, A., Biniazan, R., Hümmer, C., Kappler, S., Bayer, S., & Maier, A. (2025). **Towards Improving Mammography Quality: CNN Models for Skin Fold Positioning Error Classification**. Under internal review.

Abstracts & Clinical Validation Studies

- Bhandary Panambur, A., Sakr, K. M., Nguyen, T.-T., Maier, A., & Bayer, S. (2026). **Explainable Vision-Language Models for Mammography Report Generation**. In *BAIOSPHERE Medical 2026 Book of Abstracts*, Erlangen, Germany.
- Bhandary Panambur, A., Nguyen, T.-T., Bayer, S., & Maier, A. (2026). **LA-CLIP: Lesion-Aware Vision-Language Pretraining for Mammography via ROI-Guided Contrastive Learning**. Accepted at ECR 2026.
- Bhandary Panambur, A., Bayer, S., Maier, A., Du X., Huang, J., Lui, S. (2024). **Assessment of AI Performance in Detecting Breast Lesions on MRI: A Retrospective Study in a Chinese Cohort**. Insights into Imaging. ECR 2025.
- Bhandary Panambur, A., Hoang Dinh, A., Chung, L., Le, L. T., Maier, A., Rodriguez Ruiz, A., Andrews, M., Schmitt, B., & Bayer, S. (2024). **AI-Assisted Detection of Malignant Breast Cancer in High Breast Density Cohorts: A Retrospective Comparative Study**. Insights into Imaging. ECR 2025.
- Bhandary Panambur, A., Hoang Dinh, A., Chung, L., Le, L. T., Maier, A., Rodriguez Ruiz, A., Andrews, M., Schmitt, B., & Bayer, S. (2024). **Enhancing mammography screening sensitivity with AI-assistance: Evidence from a Vietnamese study cohort**. Insights into Imaging. ECR 2024. [↗](#)

Other Publications:

- Buess, Lukas, et al. "ALO: Addressing Class Imbalance in Radiology Report Generation through Anatomy-Level Oversampling." *Medical Imaging with Deep Learning*. 2026.
- Bhat, S., Mansoor, A., Georgescu, B., Zinnen, M., Sahu, P., Panambur, A.B., Ghesu, F.C., Grbic, S. and Maier, A., 2026. PatchCLIP enables region specific contrastive health record and image joint training with patch embedding loss: S. Bhat et al. *Scientific Reports*, 16(1), p.14688.
- Panambur AB, Madhu P, Maier A. **Classification of Luminal Subtypes in Full Mammogram Images Using Transfer Learning**. arXiv preprint arXiv:2301.09282. 2023 Jan 23. [↗](#)
- Bhat, S., Georgescu, B., Bhandary Panambur, A., Zinnen, M., Nguyen, T.-T., Mansoor, A., Elbarbary, K.K., Bayer, S., Ghesu, F.-C., Grbic, S. & Maier, A., 2025. **Exemplar Med-DETR: Toward Generalized and Robust Lesion Detection in Mammogram Images and beyond**. MICCAI 2025, S Korea, 2025. [↗](#)
- Elbarbary, K., Bhandary Panambur, A., Bhat, S., Bayer, S. & Maier, A., 2025. **MM-DETR: Emulating the Diagnostic Clinical Workflow in Multi-view Multi-modal Mammography Mass Detection**. Deep-Brea3th Workshop, MICCAI 2025, S Korea, 2025.
- Ni, Y., Bhandary Panambur, A., Liu, C., Nguyen, T.-T., Bayer, S., & Maier, A. (2026). **Opportunistic Breast Cancer Risk Stratification from Low-Dose Chest CT Using Multiple Instance Learning**. German Conference on Medical Image Computing (BVM 2026).
- Yu, Hui, Frank Forster, Adarsh Bhandary Panambur, Anna Merino, Andreas Maier, and Gaby Marquardt. **Unstained White Blood Cell Classification Using Deep Learning**. International Journal of Laboratory Hematology, vol. 45, 2023, pp. 132-133. <https://doi.org/10.1111/ijlh.14149>. [↗](#)
- Bhat S, Mansoor A, Georgescu B, Panambur AB, Ghesu FC, Islam S, Packhäuser K, Rodríguez-Salas D, Grbic S, Maier A. **AUCReshaping: improved sensitivity at high-specificity**. Scientific Reports. 2023 Nov;13(1):21097. [↗](#)
- Panambur AB, Hager F, Kuth R. **Unlocking a Medical Device**. European Patent EP 3644550 A1. 2020 Apr 29. [↗](#)